BIOL-1 Practice Test 04 — Answer Sheet

College of the Redwoods — BIOL-1: General Biology (Pelican Bay, Fall 2026) | Comprehensive Final Review (Modules 01–16)

Name:		Date:		Score:	

✂️ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (75 questions) and Part B: Fill in the Blank (30 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (75 questions)

1.		**26.**		**51.**	
2.		**27.**		**52.**	
3.		**28.**		**53.**	
4.		**29.**		**54.**	
5.		**30.**		**55.**	
6.		**31.**		**56.**	
7.		**32.**		**57.**	
8.		**33.**		**58.**	
9.		**34.**		**59.**	
10.		**35.**		**60.**	
11.		**36.**		**61.**	
12.		**37.**		**62.**	
13.		**38.**		**63.**	
14.		**39.**		**64.**	
15.		**40.**		**65.**	
16.		**41.**		**66.**	
17.		**42.**		**67.**	
18.		**43.**		**68.**	
19.		**44.**		**69.**	
20.		**45.**		**70.**	
21.		**46.**		**71.**	
22.		**47.**		**72.**	
23.		**48.**		**73.**	
24.		**49.**		**74.**	
25.		**50.**		**75.**	

BIOL-1 Practice Test 04 — Part B: Fill in the Blank (30 questions)

Fill-in answer lines. Record every blank on this page.

Part B: Fill in the Blank (30 questions)

1.		**16.**	
2.		**17.**	
3.		**18.**	
4.		**19.**	
5.		**20.**	
6.		**21.**	
7.		**22.**	
8.		**23.**	
9.		**24.**	
10.		**25.**	
11.		**26.**	
12.		**27.**	
13.		**28.**	
14.		**29.**	
15.		**30.**	

BIOL-1 Practice Test 04 — Part C: Free Response (32 questions)

Free response. Write clearly in the lined spaces.

1. **(Module 01)** State two characteristics of life and **briefly** explain each in your own words.
2. **(Module 01)** What is the difference between a **scientific theory** and a **hypothesis** in everyday lab language?
3. **(Module 02)** Explain how water's polarity helps it dissolve **ionic** compounds such as table salt at the molecular level (big picture, no deep math).
4. **(Module 02)** Why does a **buffer** matter in blood and cellular chemistry (one or two sentences of "what it does")?
5. **(Module 03)** Name the **four** major classes of biological macromolecules and, for **one** class you choose, name its monomer type.
6. **(Module 03)** What happens to a protein's function when it is **denatured**, and what is one common cause of denaturation?
7. **(Module 04)** Compare **prokaryotic** and **eukaryotic** cells in **two** sentences (nucleus + relative complexity).
8. **(Module 04)** What is the main job of the **mitochondria**, and where in the cell does it do that work?
9. **(Module 05)** Contrast **passive** and **active** transport across a membrane with **one example** of each.
10. **(Module 05)** In plain language, what happens to a cell placed in a **very salty** external solution if the membrane is permeable to water only (animal cell framing)?
11. **(Module 06)** Write one sentence linking **photosynthesis** and **cellular respiration** as complementary carbon/energy themes on Earth.
12. **(Module 06)** Why do enzymes stop speeding a reaction if the temperature becomes **too high**?
13. **(Module 07)** Describe **transcription** and **translation** in **two** short sentences, using the words **DNA**, **RNA**, **ribosome**, and **protein** sensibly.
14. **(Module 07)** Why is **DNA replication** described as **semiconservative** at the big-picture level?

15. **(Module 08)** Contrast the **goal** of mitosis versus meiosis in a multicellular animal (what each type produces).
16. **(Module 08)** Name **two** ways meiosis increases **genetic variation** besides random fertilization.
17. **(Module 09)** In a monohybrid cross **Aa** × **Aa** with complete dominance, what are the expected **phenotype** ratios in offspring?
18. **(Module 09)** Define **heterozygous** and give a **letter example** (any gene).
19. **(Module 10)** In one short paragraph, explain how **epigenetics** differs from a **mutation**.
20. **(Module 10)** Give one **realistic** example of how environment **without** changing DNA letters might still influence **trait expression** at the gene-regulation level.
21. **(Module 11)** In two sentences, what is a **GMO** in food/ag contexts, and why do people disagree ethically?
22. **(Module 11)** What does **gel electrophoresis** show a biologist about DNA samples (smaller vs larger fragments)?
23. **(Module 12)** Explain **natural selection** in four short sentences aimed at a friend who has never taken biology.
24. **(Module 12)** Name **two** independent lines of **evidence** that life shares common ancestry and **briefly** say what each shows.
25. **(Module 13)** Define **microevolution** and give **one** example of a mechanism that can change allele frequencies **other** than natural selection.
26. **(Module 13)** Contrast **directional**, **stabilizing**, and **disruptive** selection in one sentence **each**.
27. **(Module 14)** Contrast **allopatric** and **sympatric** speciation with **one sentence each**.
28. **(Module 14)** Give **one** example of a **prezygotic** barrier and **one** of a **postzygotic** barrier.
29. **(Module 15)** Contrast **exponential** vs **logistic** population growth; mention **carrying capacity** in your answer.
30. **(Module 15)** Explain the **10% energy rule** and **why** long food chains support fewer top predators than short chains from the same plant base.
31. **(Module 16)** In one paragraph, connect **DNA or genes**, **natural selection**, and **ecosystem limits** in a single biological system.

32. **(Module 16)** Explain why a strong biology answer often needs to name the **scale** of the evidence: molecule, cell, organism, population, community, or ecosystem.

Response 1 — Question Number: ______

Response 2 — Question Number: ______

Response 3 — Question Number: ______

Response 4 — Question Number: ______

Response 5 — Question Number: ______

Response 6 — Question Number: ______

Response 7 — Question Number: ______

Response 8 — Question Number: ______

Response 9 — Question Number: ______

Response 10 — Question Number: ______

Response 11 — Question Number: ______

Response 12 — Question Number: ______

Response 13 — Question Number: ______

Response 14 — Question Number: ______

Response 15 — Question Number: ______

Response 16 — Question Number: ______

Response 17 — Question Number: ______

Response 18 — Question Number: ______

Response 19 — Question Number: ______

Response 20 — Question Number: ______

Response 21 — Question Number: ______

Response 22 — Question Number: ______

Response 23 — Question Number: ______

Response 24 — Question Number: ______

Response 25 — Question Number: ______

Response 26 — Question Number: ______

Response 27 — Question Number: ______

Response 28 — Question Number: ______

Response 29 — Question Number: ______

Response 30 — Question Number: ______

Response 31 — Question Number: ______

Response 32 — Question Number: ______

BIOL-1 Practice Test 04 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (75 questions)

Choose the best answer for each question.

Module 01: Study of Life

1. Which property is shared by every living organism in this course's working definition of life?
    - A) They must move from place to place
    - B) They all use photosynthesis
    - C) They are composed of one or more cells
    - D) They are all multicellular
2. In the scientific method, a testable, proposed explanation that follows from observations is best called a:
    - A) Theory
    - B) Hypothesis
    - C) Law
    - D) Conclusion
3. In a controlled experiment, the factor the researcher deliberately changes is usually the:
    - A) Dependent variable
    - B) Confounding variable
    - C) Independent variable
    - D) Control group only
4. The maintenance of a stable internal environment (for example, temperature or blood sugar) despite external change is called:
    - A) Differentiation

- ◦ B) Homeostasis
- ◦ C) Metabolism
- ◦ D) Evolution

5. The simplest level in the hierarchy “cell → tissue → organ → organ system → organism” is the:

   - ◦ A) Cell
   - ◦ B) Organ
   - ◦ C) Tissue
   - ◦ D) Population

Module 02: Basic Chemistry

1. The **atomic number** of an element equals the number of:

   - ◦ A) Neutrons only
   - ◦ B) Electrons only in ions
   - ◦ C) Protons in the nucleus
   - ◦ D) Isotopes

2. Atoms of the same element that differ in neutron number are called:

   - ◦ A) Ions
   - ◦ B) Molecules
   - ◦ C) Isotopes
   - ◦ D) Polymers

3. A bond formed mainly by **transfer** of electrons from one atom to another, producing attraction between opposite charges, is:

   - ◦ A) Covalent
   - ◦ B) Hydrogen
   - ◦ C) Van der Waals
   - ◦ D) Ionic

4. Water is polar mainly because:
    - A) It has hydrogen atoms only
    - B) It cannot hydrogen-bond
    - C) Shared electrons are pulled unevenly, creating partial charges
    - D) It always has pH 7
5. A solution with a **lower** pH than 7 is:
    - A) Basic
    - B) Acidic
    - C) Neutral
    - D) A solution with pH above 7

Module 03: Organic Molecules

1. The element that forms the backbone of biological macromolecules is:
    - A) Carbon
    - B) Sodium
    - C) Nitrogen
    - D) Iron
2. Amino acids link by **dehydration synthesis** to form proteins, releasing:
    - A) CO_2
    - B) O_2
    - C) ATP
    - D) H_2O
3. The monomer (building block) of proteins is the:
    - A) Monosaccharide
    - B) Nucleotide
    - C) Fatty acid
    - D) Amino acid

4. Starch and glycogen are storage forms of which class of macromolecule?

   ◦ A) Lipid
   ◦ B) Carbohydrate
   ◦ C) Protein
   ◦ D) Nucleic acid

5. RNA differs from DNA in several ways; one common contrast is that RNA uses the base **uracil** instead of:

   ◦ A) Thymine
   ◦ B) Adenine
   ◦ C) Cytosine
   ◦ D) Guanine

Module 04: Cells

1. A central idea of cell theory is that **new cells** arise from:

   ◦ A) Pre-existing cells
   ◦ B) Spontaneous generation from broth
   ◦ C) Only the ER
   ◦ D) Viruses

2. Prokaryotic cells lack which structure that eukaryotic cells usually have?

   ◦ A) Ribosomes
   ◦ B) DNA
   ◦ C) A membrane-bound nucleus
   ◦ D) A plasma membrane

3. Most ATP for a typical animal cell is generated in the:

   ◦ A) Nucleus
   ◦ B) Mitochondria
   ◦ C) Lysosome
   ◦ D) Centriole

4. In plant cells but not typical animal cells, you expect to find:
   - A) A cell wall outside the membrane
   - B) Mitochondria
   - C) Ribosomes
   - D) A nucleus
5. Proteins are assembled on:
   - A) Ribosomes
   - B) The Golgi apparatus
   - C) The smooth ER only
   - D) Peroxisomes only

Module 05: Membranes

1. The basic structure of the plasma membrane is described as a:
   - A) Single layer of proteins
   - B) Fluid mosaic of lipids and proteins
   - C) Static rigid wall
   - D) DNA bilayer
2. Simple diffusion moves molecules:
   - A) Only against a gradient with ATP
   - B) Only in plants
   - C) Only as ions
   - D) With their concentration gradient, no ATP required for the diffusion step
3. In osmosis, **water** moves:
   - A) Against gradients always
   - B) Toward the area with lower water (higher solute) if the membrane allows water through
   - C) Only into dead cells
   - D) Only when sodium ions move first

4. Active transport differs from simple diffusion because it:
   - A) Never uses proteins
   - B) Can move substances against their concentration gradient using cellular energy
   - C) Only moves water
   - D) Does not depend on membrane proteins
5. Which molecules typically cross a pure lipid bilayer **least** easily without help?
   - A) Small nonpolar gases
   - B) Water (tiny, though polar)
   - C) Large, charged ions without channels/carriers
   - D) Small nonpolar molecules

Module 06: Metabolism

1. The molecule often called the cell's immediate **energy currency** is:
   - A) DNA
   - B) ATP
   - C) Glucose only
   - D) mRNA
2. Enzymes speed reactions mainly by:
   - A) Lowering activation energy
   - B) Raising activation energy
   - C) Destroying products
   - D) Removing all substrates
3. The summary outcome of **aerobic cellular respiration** (big picture) is that glucose is broken down and much energy is captured as:
   - A) Only heat with no ATP
   - B) ATP (and heat), with CO_2 and water among outputs
   - C) Light
   - D) Only lipids

4. The light-capturing reactions of photosynthesis happen in chloroplast structures associated with:
    - A) Mitochondrial cristae
    - B) Lysosomes
    - C) Thylakoids / grana (chloroplast membranes)
    - D) The nucleolus
5. Fermentation is a way some cells continue **glycolysis** when:
    - A) Oxygen is abundant
    - B) Photosynthesis is maximal
    - C) Oxygen is limited, regenerating NAD^+ so glycolysis can keep running
    - D) DNA replicates

Module 07: Molecular Genetics

1. The phrase **central dogma** most often summarizes:
    - A) DNA → RNA → protein in typical gene expression
    - B) Protein → RNA → DNA
    - C) Only replication
    - D) Mendel's laws only
2. In DNA, adenine pairs (base-pairs) with:
    - A) Guanine
    - B) Cytosine
    - C) Uracil
    - D) Thymine
3. **Transcription** produces:
    - A) An RNA copy of a gene, using DNA as a template
    - B) A new DNA double helix only
    - C) A finished protein directly
    - D) Only tRNA with no mRNA

4. **Translation** reads which molecule to build a protein?
   - A) DNA directly in the nucleus only
   - B) Only rRNA with no codons
   - C) mRNA at the ribosome
   - D) Lipid bilayers
5. A change in the **DNA sequence** that may alter an encoded protein is a:
   - A) Epigenetic mark only (same textbook definition)
   - B) Mutation
   - C) Gene flow event
   - D) Bottleneck only

Module 08: Cellular Genetics

1. Somatic cells reproduce for growth and repair mainly by:
   - A) Meiosis
   - B) Fermentation
   - C) Mitosis (exact copy distribution to two daughter cells)
   - D) Translation
2. Meiosis produces gametes with:
   - A) The same chromosome number as the parent body cell
   - B) No chromosomes
   - C) Only mitochondrial DNA
   - D) Half the chromosome number (haploid from diploid), with genetic shuffling
3. Crossing over (exchange between homologous chromosomes) increases variation mainly during:
   - A) Mitosis only
   - B) Translation
   - C) DNA replication only in skin
   - D) Prophase I of meiosis

4. A diploid human body cell has **46** chromosomes; a normal human sperm has about:

    - A) 46
    - B) 23
    - C) 92
    - D) 0

5. The term **somatic** cell refers to:

    - A) Body cells that are not gametes
    - B) Only gametes
    - C) Bacterial cells
    - D) Dead cells only

Module 09: Inheritance Genetics

1. An organism with two identical alleles for a gene is:

    - A) Heterozygous
    - B) Homozygous
    - C) Polyploid only
    - D) Always dominant

2. In simple dominance, if **T** is tall and **t** is short, a **Tt** plant's phenotype is usually:

    - A) Short
    - B) Medium always
    - C) Lethal
    - D) Tall (dominant trait expressed)

3. A **Punnett square** is used to predict:

    - A) Offspring genotypes / phenotypes from parental gametes
    - B) Mitotic spindle angles only
    - C) Only mutation rates
    - D) Ecological carrying capacity

4. A pedigree chart tracks:
    ◦ A) Only fossils
    ◦ B) Traits through family generations
    ◦ C) Only food webs
    ◦ D) Only soil nutrient cycles
5. Mendel's idea that "heritable factors" (alleles) **segregate** into gametes underpins:
    ◦ A) Only natural selection
    ◦ B) The law of segregation (intro level)
    ◦ C) Only photosynthesis
    ◦ D) Binomial nomenclature only

Module 10: Epigenetics

1. **Epigenetics** in this course most closely means:
    ◦ A) Changing the DNA letter sequence permanently by CRISPR only
    ◦ B) Random gamete fusion
    ◦ C) Fermentation pathways
    ◦ D) Regulation of gene activity without changing the DNA sequence
2. **DNA methylation** that silences a gene region is a textbook example of:
    ◦ A) A missense mutation
    ◦ B) Mitosis
    ◦ C) An epigenetic mark (often "off")
    ◦ D) Independent assortment
3. X-inactivation in female mammals (one X compacted in each cell) helps explain, for example:
    ◦ A) Why men have Y in every liver cell only
    ◦ B) Why plants lack DNA
    ◦ C) Bacterial transformation only
    ◦ D) Calico cat patch color patterns (classic intro story)

4. Compared with a **mutation**, a typical epigenetic change to gene activity:

   ◦ A) Always changes the codon sequence
   ◦ B) Cannot respond to environment
   ◦ C) Is often reversible at the “volume knob” level of expression
   ◦ D) Only happens in viruses

5. **Gene expression** (big picture) refers to:

   ◦ A) Only DNA replication
   ◦ B) A gene being used to produce RNA and often protein
   ◦ C) Only photosynthesis
   ◦ D) Only drift

Module 11: Genomics & Biotechnology

1. **PCR** is used in teaching examples to:

   ◦ A) Amplify a targeted DNA segment in a tube
   ◦ B) Photograph mice only
   ◦ C) Replace meiosis
   ◦ D) Create ATP from sunlight

2. Gel electrophoresis separates DNA fragments mainly because:

   ◦ A) The phosphate backbone makes DNA negatively charged; fragments move through a gel toward positive
   ◦ B) Heat destroys all DNA the same way
   ◦ C) Lipids are charged like DNA
   ◦ D) Only RNA enters the gel

3. A **transgenic** organism carries:

   ◦ A) No DNA
   ◦ B) Only mitochondrial DNA from mother
   ◦ C) Foreign DNA introduced into its genome (GMO framing)
   ◦ D) Only fossil genes

4. **CRISPR** is often introduced as a tool to:

    ◦ A) Edit DNA at targeted sites (with big ethics questions)
    ◦ B) Ignore genes
    ◦ C) Make fossils
    ◦ D) Measure pH only

5. A **restriction enzyme** cuts DNA at:

    ◦ A) Random weather patterns
    ◦ B) Only RNA
    ◦ C) Only proteins
    ◦ D) Specific recognition sequences

Module 12: Darwin & Evolution

1. **Natural selection** requires heritable variation, differential reproduction, and:

    ◦ A) Goal-directed "wants" of species
    ◦ B) Lamarckian inheritance of acquired traits as the main mechanism
    ◦ C) Perfect environments only
    ◦ D) Overproduction of offspring relative to resources (broad textbook framing)

2. **Descent with modification** means:

    ◦ A) Each species is fixed forever
    ◦ B) Only bacteria evolve
    ◦ C) Lineages change over time; related species share ancestors with later differences
    ◦ D) Evolution cannot be tested

3. **Homology** (for example similar limb bone layout, different functions) supports:

    ◦ A) Identical environments only
    ◦ B) No shared history
    ◦ C) Common ancestry with adaptive change
    ◦ D) Only convergent evolution in every case

4. **Fitness** in evolution means roughly:

    ◦ A) Reproductive success / allele contribution to the next generation
    ◦ B) Bench press strength only
    ◦ C) Survival with zero offspring forever
    ◦ D) Size only

5. Darwin emphasized that **populations** evolve rather than individuals mainly because:

    ◦ A) Individuals rewrite DNA by thinking
    ◦ B) Fossils are not real
    ◦ C) Allele frequencies change across generations in a population
    ◦ D) Only one individual matters per species

Module 13: How Populations Evolve

1. **Microevolution** is defined in this course as:

    ◦ A) An individual growing taller in one year
    ◦ B) Instant new phylum in one birth
    ◦ C) Only mutation in skin cells with no offspring
    ◦ D) Change in allele frequencies in a population over time

2. The **Hardy–Weinberg** model is mainly useful as:

    ◦ A) A null / baseline to compare real allele-frequency change against
    ◦ B) Proof that evolution always speeds up
    ◦ C) A ban on mutation
    ◦ D) A species concept

3. The **founder effect** is a type of:

    ◦ A) Directional selection only
    ◦ B) Gene flow from infinite migrants always
    ◦ C) Genetic drift involving a small founding group
    ◦ D) Photosynthesis error

4. A **bottleneck** reduces allele diversity mainly because:
    - A) Selection cannot happen
    - B) Mutation stops
    - C) K always equals zero
    - D) A sudden severe population crash makes survivors' alleles unrepresentative by chance
5. **Stabilizing selection** on a trait tends to:
    - A) Favor both extremes
    - B) Always eliminate all variation instantly
    - C) Require polyploidy
    - D) Favor the middle; select against extremes

Module 14: Macroevolution

1. The **biological species concept** emphasizes:
    - A) Reproductive isolation between groups in nature (big-picture framing)
    - B) Color only
    - C) Cell size only
    - D) Fossil size only
2. Different firefly flash patterns that reduce mating attempts are best described as:
    - A) A postzygotic barrier (hybrid infertility)
    - B) Geographic separation with no mating cues
    - C) Sympatric polyploidy in one generation
    - D) A prezygotic behavioral barrier to reproduction
3. Mule sterility (horse × donkey hybrid) is a classic example of:
    - A) Prezygotic temporal isolation between species
    - B) A postzygotic barrier (reduced hybrid fertility)
    - C) Convergent evolution in ungulates
    - D) A prezygotic gametic barrier only

4. **Allopatric** speciation often begins with:

    - A) Geographic separation of formerly connected populations
    - B) Same habitat, no separation
    - C) Only sympatry by definition
    - D) Only hybrid swarms always

5. **Sympatric** speciation examples in intro courses often highlight:

    - A) Only continental drift with zero genetics
    - B) Plant polyploidy in the same region as a parent type
    - C) Only fossils
    - D) Only density independence

Module 15: Population & Systems Ecology

1. Early **exponential** population growth (unlimited-resource phase) is often **J**-shaped because:

    - A) Growth rate is near zero always
    - B) Carrying capacity is always zero
    - C) Per-capita growth can stay high while numbers are still far below limits
    - D) Only density-dependent rules apply on day one

2. **Carrying capacity (K)** in a logistic model is:

    - A) The mutation rate
    - B) Always infinite
    - C) Gene flow only
    - D) A long-run ceiling set by resources and other limits

3. A disease spreading **more easily in crowded** herds is often modeled as:

    - A) Density-dependent regulation
    - B) Density-independent
    - C) Only abiotic freeze
    - D) Carrying capacity with no biology

4. The rough **~10% rule** between trophic levels implies:
    - A) 100% of energy becomes flesh always
    - B) Only a small fraction of stored energy transfers to the next consumer level (lots lost as heat/work)
    - C) Decomposers violate thermodynamics
    - D) Producers get energy from consumers' heat cycling perfectly
5. In ecosystems, **matter** (nutrient elements) cycles while energy from sunlight typically:
    - A) Is recycled exactly like nitrogen atoms with no loss
    - B) Stays only in rocks forever with zero biology
    - C) Doubles at every trophic level
    - D) Flows through and ends largely as heat (not recycled like chemical elements in the same way energy is framed)

Part B: Fill in the Blank (30 questions)

Write a short term or phrase in each blank (usually one or two words).

Module 01

1. In a controlled experiment, a group that does **not** receive the experimental treatment (used for comparison) is often called the ____ group.
2. All living things must maintain organization, reproduce, respond to stimuli, undergo metabolism, and evolve as populations; many also grow and develop while maintaining ____.

Module 02

1. A covalent bond involves **sharing** pairs of ____ between atoms.

2. A neutral atom has equal numbers of protons and ____ (in the standard model for this course).

Module 03

1. The chemical bond that links amino acids in a chain is a ____ bond.
2. Fats and oils belong to the macromolecule class called ____.

Module 04

1. Membrane-bound sacs of digestive enzymes that break down macromolecules in many eukaryotic cells are ____
2. The control center that holds most DNA in a eukaryotic cell is the ____

Module 05

1. Transport proteins that open the door for molecules **without** using ATP for that step (along a gradient) perform ____ transport.
2. The phospholipid tails of the bilayer are ____ (hydrophilic or hydrophobic)—choose the correct term.

Module 06

1. The anaerobic portion of glucose breakdown that happens in the cytoplasm (and yields a little ATP and pyruvate) is

   ____

2. The pigment that captures light energy in plants and many photosynthetic organisms is

   ____

Module 07

1. During DNA replication, the enzyme class that **matches** incoming nucleotides to the template strand (5′→3′ synthesis on a daughter strand) is

   ____

   polymerase (family name only is fine).

2. A **three**-nucleotide mRNA code that specifies an amino acid (or stop) is a

   ____

Module 08

1. The stage of mitosis where **sister chromatids** are pulled apart toward opposite poles is

   ____

2. Homologous chromosomes pair up during

   ____

   I of meiosis.

Module 09

1. An observable trait (tall, short, etc.) is called the

   ____

2. When both alleles are visibly expressed in the heterozygote (classic AB blood-type framing), inheritance is often called

   ____

Module 10

1. Methyl groups added to DNA can often **silence** nearby genes as an

   ____

   mark (adjective form used in class).

2. Proteins that help **turn transcription on or off** near promoters are often called transcription

   ____

Module 11

1. A **plasmid** is a small, circular **extra** piece of DNA often used in bacterial ______________________________ work (one-word lab context is fine).

2. To cut and paste a gene, labs often use **restriction** enzymes and

   ____

   enzyme to seal DNA back together.

Module 12

1. Heritable differences among individuals in a population are called

   ____

2. When unrelated lineages evolve **similar** traits because of similar selection pressures (not shared recent ancestry), it is

   ____

   evolution.

Module 13

1. When chance events change allele frequencies (especially in small populations), the umbrella term is genetic

   ____

2. If **two** alleles A and a have frequencies **p** and **q** in a diploid population at Hardy–Weinberg equilibrium, the expected heterozygote frequency is ____________________________ (symbolic answer).

Module 14

1. Barriers that block mating or fertilization **before** a zygote forms are ____________________________ barriers.

2. A **reduced hybrid** survival or fertility after a zygote forms is a ____________________________ zygotic barrier category.

Module 15

1. All the organisms of one species living in an area at one time form a(n) ____________________________

2. Organisms that make organic matter from inorganic carbon (often via photosynthesis) are **primary** ____________________________ (trophic role).
